Documentation of the `buoyantBoussinesqHumidityPimpleFoam` solver and `atmTurbulentMoistureFluxHumidity` boundary condition developed for OpenFOAM

Tariq Ridwan, Adeline Montlaur, David Pino

Departament de Física
Universitat Politècnica de Catalunya
Campus del Baix Llobregat
Castelldefels, Barcelona, Spain.

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1 Boussinesq approximation

The Boussinesq approximation is valid when the variation of the density induced by the temperature change is small, as Ferziger and Peric (2020, pp.15-16) state (Ferziger et al., 2020):

> "In flows accompanied by heat transfer, the fluid properties are normally functions of temperature. The variations may be small and yet be the cause of the fluid motion. If the density variation is not large, one may treat the density as constant in the unsteady and convection terms, and treat it as variable only in the gravitational term. This is called the Boussinesq approximation. One usually assumes that the density varies linearly with temperature:
>
> $$(\rho - \rho_{ref})g = -\rho_{ref} g \beta (T - T_{ref}) \tag{1}$$
>
> where β is the coefficient of volumetric expansion. This approximation introduces errors of the order of 1% if the temperature differences are below, e.g., 2° for water and 15° for air.

The error may be more substantial when temperature differences are larger; the solution may even be qualitatively wrong."

In Equation 1, ρ and T are air density and temperature, respectively. The subscript 'ref' denotes any reference value of the parameter. As the volumetric thermal expansion coefficient of air is the inverse of its absolute temperature, i.e. $\beta = 1/T_{ref}$ (Nellis and Klein, 2020, 2008; Fletcher and Johnson, 1992), the buoyancy force can be defined as:

$$b \equiv \frac{\rho_{ref} - \rho}{\rho_{ref}} g = \frac{T - T_{ref}}{T_{ref}} g \tag{2}$$

However, due to anelastic approximation:

$$\frac{T - T_{ref}}{T_{ref}} \approx \frac{\theta - \theta_{ref}}{\theta_{ref}} \tag{3}$$

where θ is the potential temperature. Therefore, Equation 2 becomes:

$$\Rightarrow b \equiv \frac{\rho_{ref} - \rho}{\rho_{ref}} g = \frac{T - T_{ref}}{T_{ref}} g = \frac{\theta - \theta_{ref}}{\theta_{ref}} g \tag{4}$$

$$\therefore \frac{\rho}{\rho_{ref}} = 1 - \frac{T - T_{ref}}{T_{ref}} = 1 - \frac{\theta - \theta_{ref}}{\theta_{ref}} = \rho_k \tag{5}$$

where ρ_k is the kinematic density for buoyancy force (denoted as `rhok` in `buoyantBoussinesqPimpleFoam` and `buoyantBoussinesqHumidityPimpleFoam` solvers), and 'ref' denotes the magnitude of a given parameter in the surrounding environment of any air parcel. See section 2.1 for the definitions of ρ and ρ_{ref}.

2 Governing equations

The conservation equations for mass, momentum, and energy:

$$\nabla \cdot \mathbf{u} = 0 \tag{6}$$

$$\rho \left[\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) \right] = -\nabla P + \rho \nu \nabla^2 \mathbf{u} + \rho \mathbf{g} \tag{7}$$

$$\frac{\partial \theta}{\partial t} + \nabla \cdot (\mathbf{u}\theta) = \alpha \nabla^2 \theta \tag{8}$$

where $\mathbf{u}$ is the velocity vector and x, y, and z are the streamwise, spanwise, and vertical coordinates, respectively. t is the time, $k = (0,\ 0,\ 1)$ is the unitary vector in the vertical direction. ν and α are the kinematic viscosity and the thermal diffusivity, respectively.

2.1 Gov. eq. with Boussinesq approximation

Considering the constant air density in the surrounding environment of any air parcel ρ_{ref} in the unsteady and convection terms and the variable density ρ in the gravitational term in Equation 7:

$$\rho_{ref} \left[\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) \right] = -\nabla P + \rho_{ref} \nu \nabla^2 \mathbf{u} + \rho \mathbf{g} \tag{9}$$

$$\Rightarrow \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{\nabla P}{\rho_{ref}} + \nu \nabla^2 \mathbf{u} + \frac{\rho}{\rho_{ref}} \mathbf{g} \tag{10}$$

$$\Rightarrow \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{\nabla P}{\rho_{ref}} + \nu \nabla^2 \mathbf{u} + \left[1 - \frac{\theta - \theta_{ref}}{\theta_{ref}} \right] \mathbf{g} \tag{11}$$

$$\therefore \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{\nabla P}{\rho_{ref}} + \nu \nabla^2 \mathbf{u} + \Big[1 - \beta(\theta - \theta_{ref}) \Big] \mathbf{g} \tag{12}$$

where the temperature-velocity coupling is achieved through the last term of Equation 12. The buoyancy parameter, known as the thermal expansion coefficient, is $\beta = 1/\theta_{ref}$.

3 Inclusion of humidity

Specific humidity is defined as the mass of water vapor (m_v) in a unit mass of air. Air is considered the summation of dry air (m_d) and water vapor(m_v). Mathematically (de Arellano et al., 2015):

$$q = \frac{\rho_v}{\rho_v + \rho_d} \tag{13}$$

where ρ_v and ρ_d are the densities of water vapor and dry air, respectively. Specific humidity units are expressed as $\mathrm{kg}_w/\mathrm{kg}_a$ or $\mathrm{g}_w/\mathrm{kg}_a$.

The virtual potential temperature, an important parameter, accounts for both the potential temperature and the humidity as (de Arellano et al., 2015):

$$\theta_v = \theta(1 + 0.61q) \tag{14}$$

de Arellano et al. (2015) states: "Since moist air is less dense than dry air, under the same conditions of temperature and pressure, θ_v is always greater than the actual temperature, but only by a few degrees."

3.1 Gov. eq. with humidity

The transport equation of the virtual potential temperature is analogous to that of the potential temperature:

$$\frac{\partial \theta_v}{\partial t} + \nabla \cdot (\mathbf{u}\theta_v) = \alpha \nabla^2 \theta_v \tag{15}$$

The momentum equation, considering the Boussinesq approximation with the virtual potential temperature, becomes (Stull, 1988):

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{\nabla P}{\rho_{ref}} + \nu \nabla^2 \mathbf{u} + \Big[1 - \beta_v(\theta_v - \theta_{v,ref})\Big]\mathbf{g} \tag{16}$$

where the buoyancy parameter is $\beta_v = 1/\theta_{v,ref}$. Applying $\theta_v = \theta(1 + 0.61q)$ in Equation 16:

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{\nabla P}{\rho_{ref}} + \nu \nabla^2 \mathbf{u} + \Big[1 - \beta_v\Big(\theta(1 + 0.61q) - \theta_{ref}(1 + 0.61q_{ref})\Big)\Big]\mathbf{g} \tag{17}$$

$$\Rightarrow \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{\nabla P}{\rho_{ref}} + \nu \nabla^2 \mathbf{u} + \Big[1 - \beta_v(\theta - \theta_{ref}) - 0.61\beta_v(\theta q - \theta_{ref}q_{ref})\Big]\mathbf{g} \tag{18}$$

$$\Rightarrow \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{\nabla P}{\rho_{ref}} + \nu \nabla^2 \mathbf{u} + \Big[1 - \beta_v(\theta - \theta_{ref}) - \frac{0.61}{\theta_{v,ref}}(\theta q - \theta_{ref}q_{ref})\Big]\mathbf{g} \tag{19}$$

$$\therefore \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{\nabla P}{\rho_{ref}} + \nu \nabla^2 \mathbf{u} + \Big[1 - \beta_v(\theta - \theta_{ref}) - 0.61\Big(\frac{\theta}{\theta_{v,ref}}q - \frac{\theta_{ref}}{\theta_{v,ref}}q_{ref}\Big)\Big]\mathbf{g} \tag{20}$$

3.2 Approximation by Limane et al. (2018) for humidity

Limane et al. (2018) used the transport equations for the potential temperature and the specific humidity separately, with some approximations made in the momentum equation:

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) = -\frac{\nabla P}{\rho_{ref}} + \nu \nabla^2 \mathbf{u} + \Big[1 - \beta(\theta - \theta_{ref}) - \beta_M\Big(q - q_{ref}\Big)\Big]\mathbf{g} \tag{21}$$

$$\frac{\partial \theta}{\partial t} + \nabla \cdot (\mathbf{u}\theta) = \alpha \nabla^2 \theta \tag{22}$$

$$\frac{\partial q}{\partial t} + \nabla \cdot (\mathbf{u}q) = \gamma \nabla^2 q \tag{23}$$

where γ is the mass diffusivity, and β_M is the mass expansion coefficient.

3.3 Analysis of the approximation by Limane et al. (2018)

The difference between the momentum Equations 20 and 21 is indicated with different colors. The approximation by Limane et al. (2018) is visible, as they omitted some terms such as $\frac{\theta}{\theta_{v,ref}}$ and $\frac{\theta_{ref}}{\theta_{v,ref}}$, while β_v is replaced with β, and 0.61 is replaced with $\beta_M = 0.60782$.

Let's analyze the errors with real-life data of a clear-sky day at 6 AM from Cabauw, the Netherlands. For a given reference potential temperature of $\theta_{ref} = 289.19$ K and specific humidity of $q_{ref} = 0.009$ kg$_w$/kg$_a$, the virtual potential temperature $\theta_{v,ref}$ is:

$$\theta_{v,ref} = \theta_{ref}(1 + 0.61q_{ref}) = 289.19(1 + 0.61 \times 0.009) = 290.78 \text{ K} \tag{24}$$

Therefore:

$$\beta = 1/\theta_{ref} = 0.003457934 \tag{25}$$

$$\beta_v = 1/\theta_{v,ref} = 0.003439972 \tag{26}$$

$$\frac{\theta_{ref}}{\theta_{v,ref}} = 0.994531949 \approx 1.0 \tag{27}$$

with around 0.5% error.

As Ferziger et al. (2020) stated that the Boussinesq approximation introduces 1% error for a temperature difference below 15 K, therefore $\theta = \theta_{v,ref} + 15 = 305.78$ K gives:

$$\frac{\theta}{\theta_{v,ref}} = 1.0515 \tag{28}$$

which is around 5.15% error. Lastly, the β_M in Equation 21 is $\beta_M = 0.60782 \approx 0.61$, both values are fine.

Therefore, the only major source of error is $\frac{\theta}{\theta_{v,ref}}$. However, it is still expected to produce reasonable results if the temperature difference is within 15 K. For higher temperature differences, the $\frac{\theta}{\theta_{v,ref}}$ term may increase the error. To stay cautious, it is better to use Equation 20 for momentum (instead of Equation 21), Equation 22 for potential temperature, and Equation 23 for specific humidity.

3.4 Moisture flux boundary condition

The `atmTurbulentMoistureFluxHumidity` boundary condition (Ridwan, 2026) is developed along with the `buoyantBoussinesqHumidityPimpleFoam` solver to calculate the specific humidity at the boundary using the provided vertical kinematic moisture flux (Q_{kin}) data. The boundary condition expression is:

$$\nabla q = \frac{Q_{kin}}{\gamma} \tag{29}$$

where,

$$Q_{kin} = \frac{Q_E}{\rho L_v} = \frac{Q_m}{\rho} \tag{30}$$

and

$$Q_m = \frac{Q_E}{L_v} \tag{31}$$

Here, Q_E is the latent heat flux in W/m², Q_m is the moisture mass flux in kg/m²/s, and L_v is the latent heat of vaporization in J/kg. For a kinematic moisture flux of $Q_{kin} = 1.4435\text{E-}04$ kg/kg·m/s, the `atmTurbulentMoistureFluxHumidity` boundary condition in OpenFOAM can be written as:

```
wall_moisture_flux_BC
{
        type            atmTurbulentMoistureFluxHumidity;
        MoistureSource  flux;
        gammaEff        gammaEff; // from shEqn.H line 8 of the
            buoyantBoussinesqHumidityPimpleFoam solver
        value           uniform 0.00904; // initial guess of specific humidity,
            typically from initial condition
        MoistureFlux    uniform 1.4435E-04; // Kinematic moisture flux in kg/kg m/s
}
```

If the user has Q_E or Q_m data available, those can be converted to Q_{kin} using Equation 30.

4 Consideratinon for LES

In large eddy simulations, the effective kinematic viscosity ν_{eff}, effective thermal diffusivity α_{eff}, and mass diffusivity γ_{eff} are considered in the diffusion terms of the momentum equation, potential temperature, and specific humidity transport equations, respectively:

$$\nu_{eff} = \nu + \nu_t \tag{32}$$

$$\alpha_{eff} = \alpha + \alpha_t \tag{33}$$

$$\gamma_{eff} = \gamma + \gamma_t \tag{34}$$

where variables with the t subscript are the contribution of the subgrid scale (SGS) turbulence. The turbulent Prandtl number is:

$$Pr_t = \nu_t / \alpha_t \tag{35}$$

while the turbulent Schmidt number is:

$$Sc_t = \nu_t / \gamma_t \tag{36}$$

5 Implementation in OpenFOAM

In the `buoyantBoussinesqHumidityPimpleFoam` solver (Ridwan, 2026), Equation 21 for momentum, Equation 22 for potential temperature, and Equation 23 for specific humidity are implemented.

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